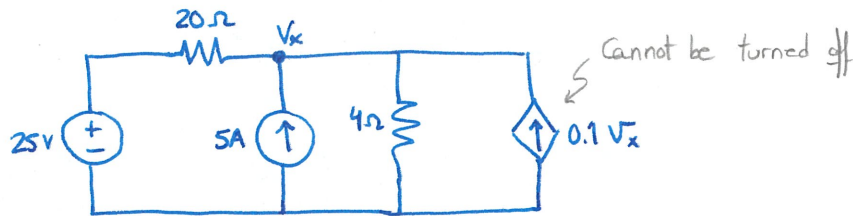
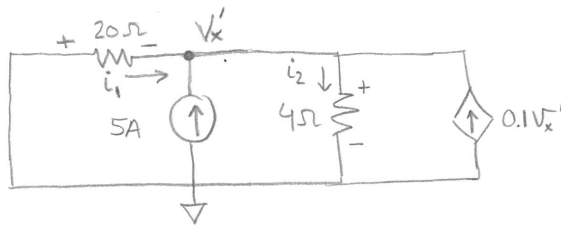


EXERCISE 2 (Superposition)

Find V_x



① Turn off voltage source ($V=0$)



Just one non-reference node $\Rightarrow$ nodal analysis

KCL @ node V'_x :

$$i_1 + 5 - i_2 + 0.1V'_x = 0$$

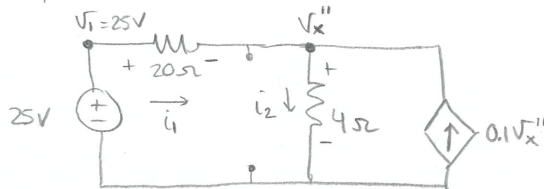
$$\frac{0 - V'_x}{20} + 5 - \frac{V'_x - 0}{4} + 0.1V'_x = 0$$

$$-\frac{V'_x}{20} + 5 - \frac{V'_x}{4} + 0.1V'_x = 0$$

$$\left(-\frac{1}{20} - \frac{1}{4} + 0.1\right)V'_x = -5$$

$$-0.2V'_x = -5 \Rightarrow V'_x = 25V$$

② Turn off current source ($i=0$)



We can easily do nodal analysis again:

KCL @ node V''_x :

$$i_1 - i_2 + 0.1V''_x = 0$$

$$\frac{25 - V''_x}{20} - \frac{V''_x}{4} + 0.1V''_x = 0$$

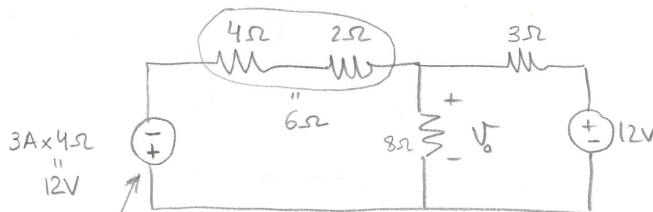
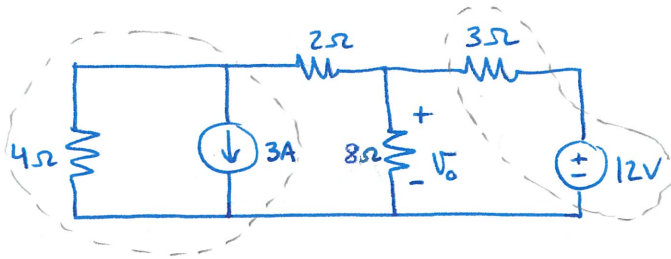
$$\frac{25}{20} - \frac{V''_x}{20} - \frac{V''_x}{4} + 0.1V''_x = 0 \Rightarrow -0.2V''_x = -\frac{25}{20} \Rightarrow V''_x = 6.25V$$

③ Calculate final value

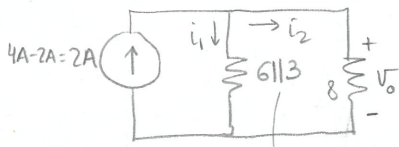
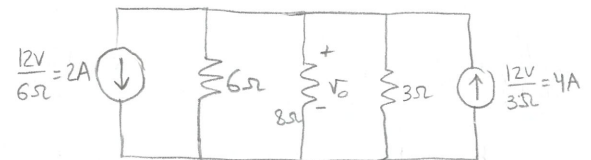
$$\underline{V_x = V'_x + V''_x = 25 + 6.25 = 31.25V}$$

EXERCISE 3 (Source transformation)

Use source transformation to find V_o



Careful! Since arrow is pointing down, the + is at the bottom



$$6 \parallel 3 = \frac{1}{\frac{1}{6} + \frac{1}{3}} = \frac{1}{\frac{1+2}{6}} = \frac{6}{3} = 2\Omega$$

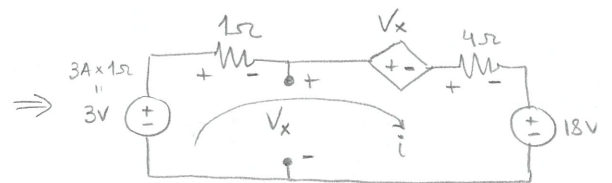
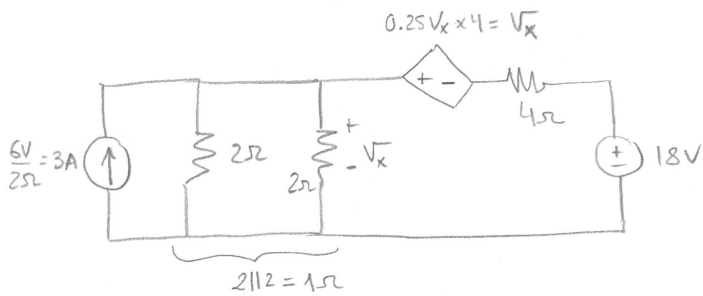
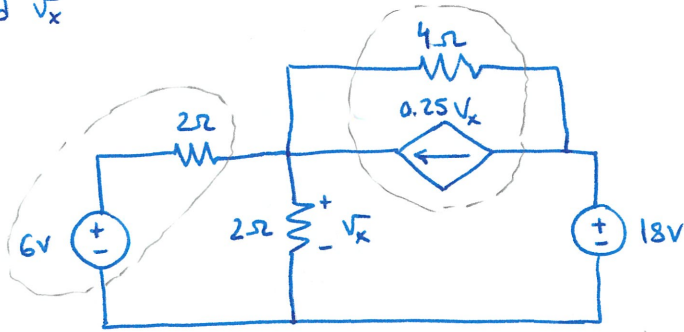
Current divider:

$$i_2 = 2 \frac{2 \parallel 8}{8} = \frac{2}{8} \times \frac{8}{5} = \frac{2}{5} \text{ A} \Rightarrow \underline{V_o = 8i_2 = \frac{16}{5} \text{ V} = 3.2 \text{ V}}$$

$$2 \parallel 8 = \frac{1}{\frac{1}{2} + \frac{1}{8}} = \frac{1}{\frac{4+1}{8}} = \frac{8}{5}$$

EXERCISE 4 (Source Transformation)

Find V_x



KVL @ big mesh:

$$-3 + 1i + V_x + 4i + 18 = 0 \quad (1)$$

KVL @ small mesh to find V_x

$$-3 + 1i + V_x = 0 \Rightarrow V_x = 3 - i \quad (2)$$

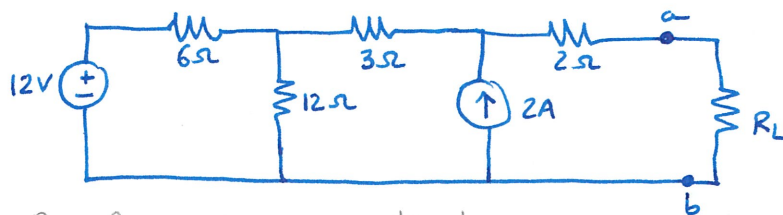
Substitute (2) into (1):

$$-3 + i + (3 - i) + 4i + 18 = 0 \Rightarrow i = -\frac{18}{4} = -4.5A$$

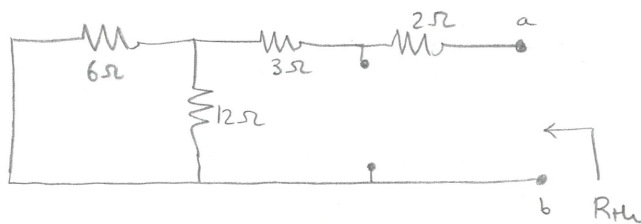
$$\underline{V_x = 3 - i = 3 - (-4.5) = 7.5V}$$

EXERCISE 10 (Maximum power transfer)

Find R_L for maximum power transfer and find the maximum power.



* $R_L = R_{th}$ for maximum power transfer



$$R_L = R_{th} = 2 + 3 + (12 \parallel 6) = 9 \Omega$$

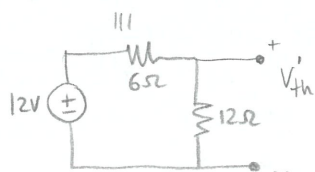
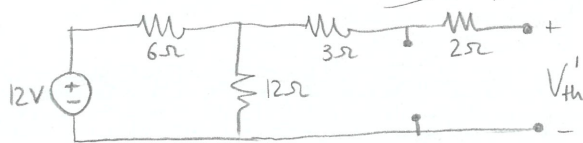
$$12 \parallel 6 = \frac{1}{\frac{1}{12} + \frac{1}{6}} = \frac{1}{\frac{1+2}{12}} = \frac{12}{3} = 4 \Omega$$

$$* P_{max} = \frac{V_{th}^2}{4R_{th}}$$

We calculate $V_{th} = V_{oc}$. We can use nodal analysis, mesh analysis, superposition, source transformation... I will use superposition.

- Turn off current source ($i=0$)

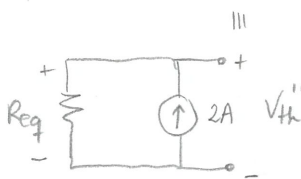
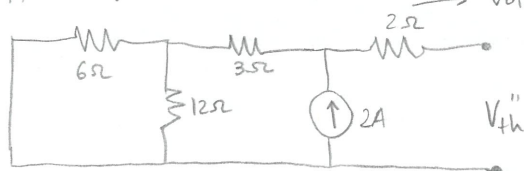
Voltage across this resistors is 0V, since there is no current flowing, due to the open circuit.



$$V'_{th} = V_{12\Omega} = 12 \frac{12}{12+6} = 8V \quad (\text{voltage divider})$$

- Turn off voltage source ($v=0$)

Voltage/current in this resistor are equal to 0, due to the open circuit



$$R_{eq} = 12 \parallel 6 + 3 = 7 \Omega$$

$$12 \parallel 6 = 4 \Omega$$

$$V''_{th} = 2A \times 7\Omega = 14V$$

$$V_{th} = V'_{th} + V''_{th} = 8 + 14 = 22V$$

$$P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{22^2}{4 \times 9} = 13.444 W$$